Video-Behavior Alignment: Session R2

Animal: 4_Milka | Date: 2023-08-16 | Pipeline: video_alignment_run_ready.py

| MAE: 5.89 s | Confidence mean: ~0.96

Pipeline Run Settings

Parameter	Value
--frame-step	10 (166,001 frames sampled)
--sweep	No (fixed params from R1)
Detector mode	Ensemble (PC1 + template)
Runtime	~2h 46min

Alignment Results

Warp Parameters

Linear temporal warp: video_time = stretch × behavior_time + offset

Metric	Value	Interpretation
Warp stretch	1.000190	Clocks nearly identical rate
Warp offset	4.25 s	Within ±5s bound (just)
Warp RMSE	32.77 s	Much better than R1 (151 s)
MAE	5.89 s	No sweep run
RMSE	12.15 s	Some larger outlier errors
Clock drift	~0.6 s over session	Very stable (R1 had 4.8 s)
Trials aligned	**207 / 209 (99%)**	2 NaN (trials 2, 8)
tp / fp / fn	2 / 0 / 0	±50ms tolerance — improving

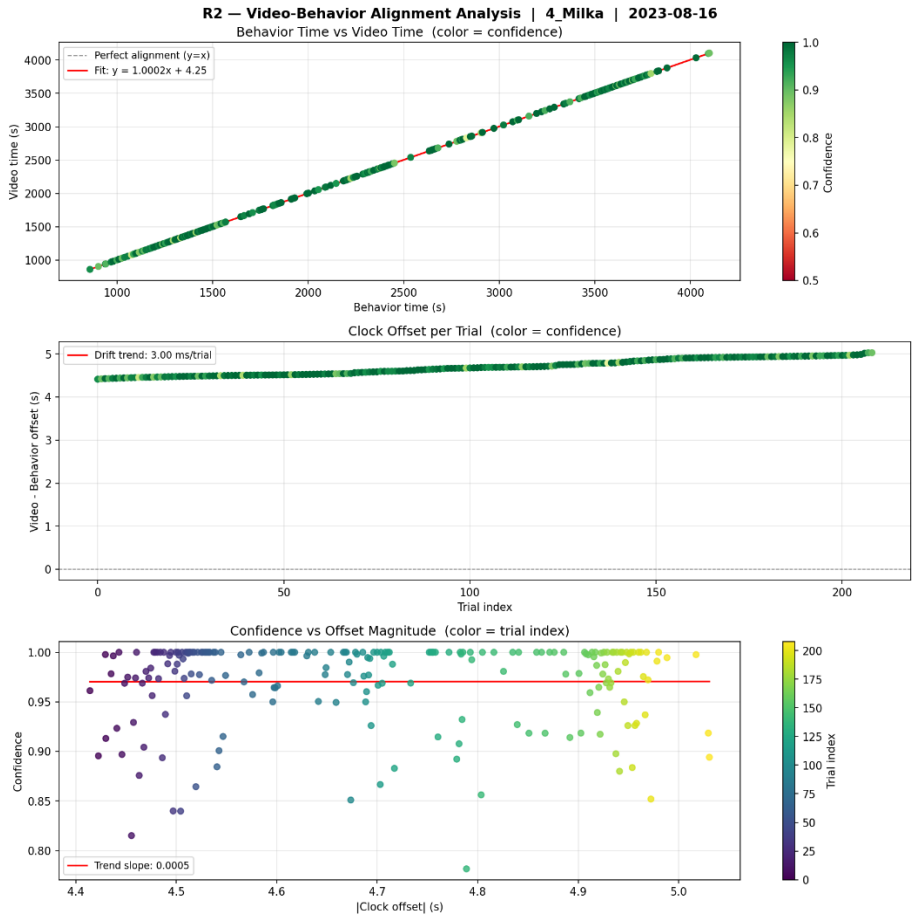
Confidence Score Distribution

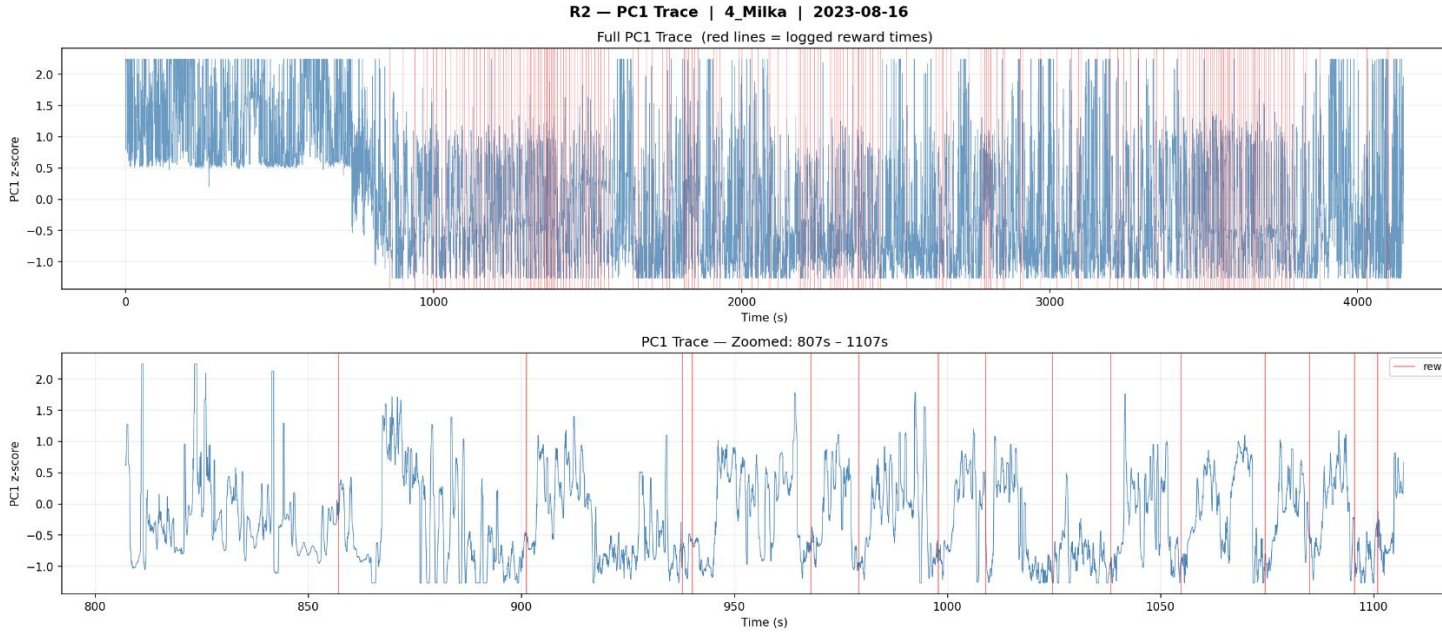
Metric	Value	vs R1
Min confidence	0.840 (trial 33)	R1 min was 0.51 ✓ better
Mean confidence	~0.960	R1 mean was ~0.80 ✓ better

Per-Trial Alignment Table (sampled)

Trial	Bhv time (s)	Video time (s)	Offset (s)	Confidence
0	857.10	861.52	+4.42	0.961
1	901.17	905.60	+4.43	0.896
2	937.77	942.20	+4.43	**0.998**
10	1054.83	1059.28	+4.45	0.975
20	1175.18	1179.65	+4.47	0.984
30	1264.46	1268.95	+4.49	0.988
40	1339.14	1343.64	+4.51	**1.000**
50	1408.00	1412.51	+4.51	0.963
70	1538.00	1542.53	+4.53	0.951
90	1661.00	1665.55	+4.55	0.972
110	1788.00	1792.56	+4.56	0.983
130	1921.00	1925.58	+4.58	0.941
150	2072.00	2076.59	+4.59	0.957
170	3541.33	3546.26	+4.93	0.978
190	3679.66	3684.61	+4.95	**0.998**
208	4099.08	4104.11	+5.03	0.894

Visualization





Plots 1,2,3,4,5. This time the second plot shows no significant correlation between confidence and offset, which may undermine the previous idea that the confidence score increasing leads immediately to lesser alignment errors . The last plot shows the pc1 is in tune with the reward near the end of the experiment but not aty the start.

Output Files

File	Contents
sync_Camera1...json	Canonical sync: warp params + video_time per trial + confidence scores
Camera1..._signal.csv	time vector + pc1_z trace (full session)
Camera1..._diagnostics.npz	NumPy: pc1_z, roi_scores, roi_coords (1.66M frames)

Known Issues

Issue	Impact	Fix
No sweep run for R2	MAE 5.89s may not be optimal	MAE likely drops with sweep
2 NaN trials (2, 8)	Missed detections	Investigate video and access if the derivative threshold is too strict